Experiment 4: Word Frequency Analysis and Visualization

```
!pip -q install wordcloud
import re
from collections import Counter
import matplotlib.pyplot as plt
from wordcloud import WordCloud

# Sample text: replace this paragraph with your lab text if needed
text = """
Natural Language Processing is a subfield of artificial intelligence.
NLP helps computers understand human language.
Natural language processing is used in chatbots,
machine translation, sentiment analysis and text summarization.
"""

# Convert to lowercase and extract words
words = re.findall(r"\b[a-z]+\b", text.lower())

# Count words
word_counts = Counter(words)
total_words = len(words)

print("Total number of words:", total_words)

if word_counts:
    max_frequency = max(word_counts.values())
    most_frequent = [
        word for word, count in word_counts.items()
        if count == max_frequency
    ]

    print("Most frequent word(s):", most_frequent)
    print("Frequency:", max_frequency)

    # Top 10 words
    top_words = word_counts.most_common(10)
    labels = [word for word, count in top_words]
    frequencies = [count for word, count in top_words]

    plt.figure(figsize=(10, 5))
    plt.bar(labels, frequencies)
    plt.title("Word Frequency (Top 10 Words)")
    plt.xlabel("Words")
    plt.ylabel("Frequency")
    plt.xticks(rotation=45, ha="right")
    plt.tight_layout()
    plt.show()
```

```
    # Word cloud
    wordcloud = WordCloud(
        width=1000,
        height=500,
        background_color="white"
    ).generate_from_frequencies(word_counts)

    plt.figure(figsize=(12, 6))
    plt.imshow(wordcloud, interpolation="bilinear")
    plt.title("Word Cloud Representation")
    plt.axis("off")
    plt.show()
else:
    print("No words found in the input text.")
```

```
Total number of words: 29
Most frequent word(s): ['language']
Frequency: 3
```



